### File Listing

View directory contents and file details.

| | |
|---|---|
| ls | List directory contents |
| ls -la | List all with details |
| ls -lh | Human readable sizes |
| ls -R | List recursively |

### Navigation

Move between directories in the filesystem.

| | |
|---|---|
| cd | Change directory |
| cd .. | Go up one level |
| cd ~ | Go to home |
| cd - | Go to previous dir |
| pwd | Print working directory |
| pushd/popd | Directory stack |

### Create & Delete

Create and remove files and directories.

| | |
|---|---|
| mkdir | Create directory |
| mkdir -p a/b/c | Create nested dirs |
| touch | Create empty file |
| rm | Remove file |
| rm -rf dir | Remove dir recursively |
| rmdir dir | Remove empty dir |

### Copy & Move

Copy, move, and link files.

| | |
|---|---|
| cp | Copy file |
| cp -r src dest | Copy directory |
| mv | Move or rename |
| ln -s | Create symlink |

### File Viewing

Display and browse file contents.

| | |
|---|---|
| cat | Display contents |
| less | View with pagination |
| head | First N lines |
| tail | Last N lines |
| tail -f file | Follow changes |

### File Search

Find files by name, type, or date.

| | |
|---|---|
| find | Find files by criteria |
| find . -name "*.t | Find by name |
| find . -type d | Find directories |
| locate | Quick search (db) |

### Text Search

Search for patterns inside files.

| | |
|---|---|
| grep | Search in file |
| grep -r "text" di | Search recursively |
| grep -i "text" | Case insensitive |
| grep -n "text" | Show line numbers |
| grep -v "text" | Invert match |

### File Info

Get information about files and commands.

| | |
|---|---|
| file | Determine file type |
| stat | Display file status |
| type | Show command type |
| which | Locate a command |
| whereis | Locate binary/source |
| basename | Strip directory path |

### Permissions

Manage file permissions and ownership.

| | |
|---|---|
| chmod | Change permissions |
| chmod 755 file | Set rwxr-xr-x |
| chmod +x file | Add execute |
| chown | Change owner |
| chgrp | Change group |
| chattr | Change attributes |
| umask | Set default perms |

### User Info

Display information about users.

| | |
|---|---|
| whoami | Current username |
| id | User ID and groups |
| who | Logged in users |
| w | Who + what they do |
| last | Login history |

### User Management

Create, modify, and delete users.

| | |
|---|---|
| useradd | Create user |
| userdel | Delete user |
| passwd user | Change password |
| usermod | Modify user |
| su | Switch user |
| sudo | Run as root |

### Group Management

Create and delete user groups.

| | |
|---|---|
| groupadd | Create group |
| groupdel | Delete group |

### Process Viewing

Monitor running processes.

| | |
|---|---|
| ps | List processes |
| ps aux | All processes |
| top | Interactive viewer |
| htop | Better viewer |
| pgrep | Find by name |
| pidof | Find PID by name |
| pstree | Process tree |

### Process Control

Terminate and manage processes.

| | |
|---|---|
| kill | Terminate process |
| kill -9 PID | Force kill |
| pkill | Kill by name |
| killall name | Kill all by name |
| timeout | Run with time limit |

### Background Jobs

Run and manage background processes.

| | |
|---|---|
| cmd & | Run in background |
| bg | Background current |
| fg | Foreground current |
| jobs | List jobs |
| nohup | Immune to hangups |
| at | Schedule command |

### Disk Usage

Check disk space and filesystem health.

| | |
|---|---|
| df | Disk space |
| du | Directory size |
| du -h --max-depth | Subdir sizes |
| [free](/post/free | Memory usage |
| fdisk | Partition table |
| fsck | Check filesystem |

### Network

Configure and test network connections.

| | |
|---|---|
| ip a | Show IP addresses |
| ifconfig | Network interfaces |
| ping | Test connectivity |
| curl | Transfer data |
| wget | Download file |
| dig | DNS lookup |

### SSH & Transfer

Secure remote access and file transfer.

| | |
|---|---|
| ssh | SSH connection |
| ssh -p 22 user@hc | Custom port |
| scp | Secure copy |
| rsync | Sync files |
| sftp | Secure FTP |
| ftp | FTP transfer |

### Network Tools

Advanced network diagnostics and scanning.

| | |
|---|---|
| `ss -tulpn` | Listening ports |
| `netstat -tulpn` | Listening ports |
| `lsof -i :80` | What uses port 80 |
| `nmap` | Network scanner |
| `nc` | Netcat utility |
| `tcpdump` | Packet capture |

### Archives

Create and extract compressed archives.

| | |
|---|---|
| `tar` | Archive files |
| `tar -czvf a.tar.g` | Create tar.gz |
| `tar -xzvf a.tar.g` | Extract tar.gz |
| `gzip` | Compress file |
| `gunzip` | Decompress file |
| `zip -r a.zip dir` | Create zip |
| `unzip a.zip` | Extract zip |

### Text Processing

Transform and manipulate text data.

| | |
|---|---|
| `sort file` | Sort lines |
| `uniq file` | Remove duplicates |
| `wc` | Count lines/words |
| `cut` | Cut columns |
| `tr` | Translate chars |
| `paste` | Merge lines |

### Sed & Awk

Stream editing and pattern processing.

| | |
|---|---|
| `sed 's/old/new/g'` | Find & replace |
| `sed -i 's/old/new` | In-place replace |
| `awk` | Pattern processing |
| `awk '{print $1}'` | Print 1st field |
| `awk -F: '{print $` | Custom delimiter |

### Pipes & Redirection

Connect commands and redirect output.

| | |
|---|---|
| `tee` | Read and write |
| `xargs` | Build arguments |
| `diff` | Compare files |
| `cmd > file` | Redirect output |
| `cmd >> file` | Append output |
| `cmd 2>&1` | Stderr to stdout |

### System Info

Display system and hardware information.

| | |
|---|---|
| `uname -a` | System info |
| `hostname` | Show hostname |
| `uptime` | System uptime |
| `date` | Current date/time |
| `dmesg` | Kernel messages |
| `history` | Command history |

### Timing & Scheduling

Schedule and time command execution.

| | |
|---|---|
| `time` | Measure runtime |
| `watch` | Run repeatedly |
| `[sleep](/post/how` | Delay execution |
| `at` | Schedule once |
| `crontab -e` | Schedule recurring |

### System Control

Manage system power and settings.

| | |
|---|---|
| `shutdown` | Shutdown system |
| `reboot` | Restart system |
| `sysctl` | Kernel parameters |
| `wall` | Broadcast message |
| `export` | Set env variable |

### Package Management

Install and manage software packages.

| | |
|---|---|
| apt | Debian packages |
| apt install pkg | Install package |
| apt update | Update package list |
| rpm | RPM packages |
| yum install pkg | RHEL/CentOS install |
| dnf install pkg | Fedora install |

### Kernel Modules

Load and manage kernel modules.

| | |
|---|---|
| lsmod | List modules |
| modprobe | Load module |
| rmmod | Remove module |

### Bash Built-ins

Common shell built-in commands.

| | |
|---|---|
| echo | Print text |
| source | Execute script |
| read | Read input |
| printf | Format output |
| exit | Exit shell |
| wait | Wait for process |

### File Operations

Rename, delete, and edit files.

| | |
|---|---|
| rename | Batch rename |
| unlink | Remove link |
| nano | Text editor |